DMBI Module 5

Explain decision tree classifier.

A Decision Tree is a supervised machine learning algorithm used for both classification and regression tasks (though commonly for classification). It models decisions and their possible consequences as a tree-like structure.

- Root Node: Represents the entire dataset and the best feature to split on first.
- Internal Nodes (Decision Nodes): Represent a feature (attribute) with branches leading to possible values.
- Branches: Represent the outcome of a test (e.g., `Age < 30`).
- Leaf Nodes (Terminal Nodes): Represent a class label (decision).

It recursively partitions the data based on feature values, selecting the most informative feature at each step until all data in a subset belongs to the same class or a stopping criterion is met.

Steps for ID3 algorithm with example.

Decision Tree Induction and Attribute Selection

Induction:

- Definition: A supervised learning method that builds a tree-like model to make decisions based on data features.
- Process: Recursively partitions the dataset by selecting the best attribute at each node.
- Split Criteria: Uses metrics like *Information Gain*, *Gini Index*, or *Gain Ratio* to choose the best attribute for splitting.
- Nodes:
    - Root Node: Topmost node containing the entire dataset.
    - Internal Nodes: Represent tests on attributes (e.g., `Age > 30?`).
    - Leaf Nodes: Represent the final class label or prediction.
- Stopping Rules: Stops when all data is classified, max depth is reached, or no further gain is possible.
- Advantages: Easy to interpret, handles both numeric/categorical data, and requires little pre-processing.

- Disadvantages: Prone to overfitting; small data changes can alter the tree significantly.
- Solutions: Uses pruning (removing weak branches) to improve generalization.
- Algorithms: Includes `ID3`, `C4.5`, and `CART`.

Attribute Selection:

- Attribute selection identifies which feature best splits the data at each node. The goal is to select the attribute that creates the purest child nodes (nodes with mostly one class).

Explain the concept of information gain and Gini value used in the decision tree algorithm.

1. Information Gain

- Based on: Entropy (measure of randomness or impurity).
- Goal: Measures the reduction in uncertainty after a split.
- Selection Criterion: Higher Information Gain → Better attribute.
- Used in Algorithms: `ID3`, `C4.5`.
- Formula Concept:

$$IG = Entropy(before) - Entropy(after)$$

- Example: Splitting data by "Outlook" significantly reduces confusion about "Play/No Play" → High Information Gain.

2. Gini Index

- Based on: Impurity of a node.
- Goal: Measures how often a randomly chosen element would be misclassified.
- Selection Criterion: Lower Gini Value → Better split `(0 = Pure, 1 = Impure)`.
- Used in Algorithms: CART (Classification and Regression Trees).
- Formula Concept:

$$Gini = 1 - \sum (p_i)^2$$

  (Where (`p_i`) is the probability of a class)
- Example: A node containing mostly "`Yes`" results in a very low Gini value, indicating a clean split.

Naïve Bayes Classifier

Definition: A probabilistic machine learning algorithm based on Bayes' Theorem with a strong "`naïve`" assumption that all features are independent of each other given the class label.

Bayes' Theorem:

$$P(y|X) = \frac{P(X|y) \cdot P(y)}{P(X)}$$

Where:

- (P(y | X)) = Posterior probability of class (`y`) given features (`X`)
- (P(X | y)) = Likelihood (probability of features given class)
- (P(y)) = Prior probability of class
- (P(X)) = Evidence (normalizing constant)

The "Naïve" Assumption:

$$P(X|y) = P(x_1|y) \times P(x_2|y) \times \ldots \times P(x_n|y)$$

Advantages:

- Works well with high-dimensional data (e.g., text classification)
- Requires small training data
- Very fast and scalable
  Disadvantages:
- Feature independence assumption is often unrealistic
- Poor performance with correlated features
  Applications: Spam filtering, sentiment analysis, document classification.

Simple Linear Regression

Definition: A statistical method that models the relationship between a single independent variable (`X`) and a dependent variable (Y `)` using a straight line.

Equation:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

Where:

- (Y) = Dependent variable (output / prediction)
- (X) = Independent variable (input / feature)

- (\beta_0) = Intercept (value of `Y` when `X` = 0)
- (\beta_1) = Slope (change in `Y` per unit change in X `)`
- (\epsilon) = Error term (random noise)

Formula for Slope:

$$\beta_1 = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{\sum(X_i - \bar{X})^2}$$

Assumptions:

1. Linear relationship between `X` and `Y`
2. Independence of errors
3. Homoscedasticity (constant variance of errors)
4. Normality of errors

Example: Predicting house price (`Y`) based on house size in sq. ft. (`X`).
Evaluation Metric: Mean Squared Error (`MSE`) or R-squared ((`R^2`)).